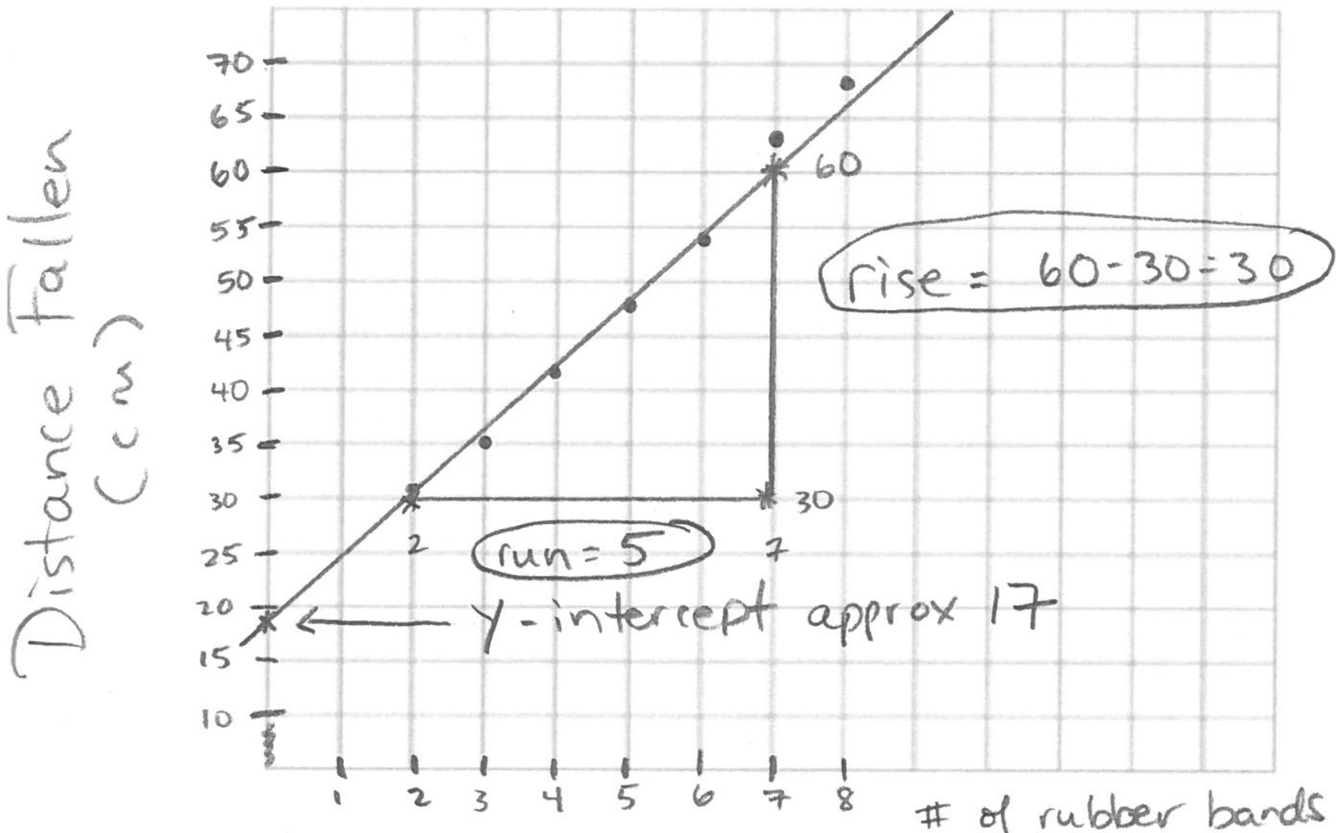
Answer Key

The Barbie Bungee Jump – INDIVIDUAL COMPONENT /15

6. **Mr. Heptinstott decided to try this investigation himself. Using slightly smaller rubber bands, he came up with the following data:**

Number of rubber bands	2	3	4	5	6	7	8
Distance fallen (cm)	31	35	42	48	54	63	68

a) Graph the scatter plot and find the line of best fit /4



b) Find the gradient of the line of best fit /2

$\text{gradient} = \frac{\text{rise}}{\text{run}} = \frac{30}{5} = 6$ (approx)

c) Find the y-intercept of the line of best fit /2

Approx. 17

d) Find the equation of the line of best fit /3

$y = mx + c$; $m = 6$, $c = 17$; $y = 6x + 17$ – allow for some variation.

e) If Mr. Heptinstott plans to drop his Barbie from a height of 10.4m, how many rubber bands will he need to attach to his Barbie to come as close to the ground without hitting it? /4

10.4 m = 1040 cm ; $y = 6x + 17$

$1040 - 17 = 6x + \cancel{17} - \cancel{17}$; $\frac{1023}{6} = \frac{\cancel{6}x}{\cancel{6}}$ ÷ 6 ; $x = 170.5$ rubber bands (approx).